

XIAOZHEN WU

✉ wxzher@outlook.com · 🌐 <https://github.com/wxzher> · 🔗 <https://wxzher.github.io/>

🎓 EDUCATION

Wuhan University | Wuhan, China

Sep 2024 – Jun 2027

Master of Arts in Intelligent Communication

- **GPA & Academic Performance:** 3.66/4.0, weighted average score: 91.94/100
- **Selected Coursework:** Social Network Computation, Optimization Theory and Methods, Operations Research, Linear Algebra, Advanced Mathematics, Modern AI Algorithms.
- **Honors:** First-class Graduate Academic Scholarship (2025), National First Prize in the 22nd "Huawei Cup" Chinese Graduate Mathematical Modeling Competition.

Wuhan University | Wuhan, China

Sep 2020 – Jun 2024

Bachelor of Arts in Network Communication

- **GPA:** 3.92/4.00 (Major Rank: 1/42)
- **Honors:** "Outstanding Student" Title (2020–2023), National Excellence Award, 14th Blue Bridge Cup Web Application Development Competition.

📄 PUBLICATIONS

- Zhang, M., Dong, Q., & **Wu, X.** *How misinformation diffuses on online social networks: radical opinions, adaptive relationship and algorithmic intervention.* IEEE Transactions on Computational Social Systems (SCI Q1).
 - Designed dynamic models to capture the co-evolution of users' misinformation beliefs and social relations under algorithmic interventions.
 - Simulated the impact of five content distribution mechanisms on social network structures.
- Dong, Q., Zhang, M., Wu, S., & **Wu, X.** *EMIF: Evidence-aware Multi-source Information Fusion Network for Explainable Fake News Detection.*
 - Proposed a co-attention network to capture semantic conflicts between comments and news articles, enhancing cross-source evidence consistency.

🏛️ RESEARCH EXPERIENCE

Platform Swinging Dynamics & Opinion Evolution Modeling

Current

Independent Researcher Wuhan University

- Developing a data-driven simulation framework utilizing Graph Neural Network (GNN) and Large Language Model (LLM) architecture to model cross-platform discourse shifts.
- Integrating Natural Language Processing (NLP) with Agent-Based Modeling (ABM) to parse semantic nuances, adopting the competitive table tennis community as the primary cross-platform dataset for empirical simulation.

High-Speed Train Bearing Cross-Domain Fault Diagnosis

Sep 2025

Core Developer Academic Project

- Designed a diagnostic framework combining deep transfer learning and multi-dimensional explainability using ResNet-18, achieving F1-score > 99.9% on the source domain dataset.
- Implemented Unsupervised Domain Adaptation (UDA) via Domain Adversarial Neural Networks (DANN), coupled with high-confidence pseudo-labels to mitigate cross-domain distribution discrepancy.

Mechanism of Misinformation Diffusion in Multilayer Social Networks Jul 2022 – Aug 2023

Core Team Member Provincial College Students' Innovation and Entrepreneurship Training Program

- Investigated cross-platform misinformation diffusion mechanisms and formalized coupled correspondence rules for multilayer social networks.
- Constructed scale-free and community-coupled network simulation environments on NetLogo, and ran quantitative diffusion experiments via the BehaviorSpace module.

PROFESSIONAL EXPERIENCE

iFlytek Healthcare Co., Ltd. | Wuhan, China

May 2025 – Sep 2025

Assistant Research Algorithm Engineer

- **Epidemiological Simulation & Complex Dynamics:** Developed an Agent-Based Modeling (ABM) simulation system alongside 7 compartmental models including Susceptible-Infected-Recovered (SIR) and Modified Susceptible-Exposed-Infected-Recovered (MSEIR) to quantify policy intervention efficacy. Restructured ordinary differential equations (ODEs) to dynamically compute basic reproduction number R_0 for precise epidemic trend forecasting.
- **Named Entity Recognition (NER) System Optimization:** Rebuilt the medical address recognition module based on Bidirectional Long Short-Term Memory-Conditional Random Field (Bi-LSTM-CRF) architecture for automatic geographic information extraction. Resolved textual ambiguity through bad-case attribution and incremental model retraining.
- **Multi-Task Learning (MTL) Joint Model:** Built an MTL framework to simultaneously execute food category classification and contaminated diet detection. Expanded training corpus by generating 6,000+ augmented samples via LLM assistance and introduced Focal Loss, lifting overall F1-score from 76% to 94%.
- **Spatiotemporal Analysis & Prediction:** Constructed a teacher-student heterogeneous network regularized by KL divergence and graph Laplacian for cross-domain knowledge transfer. Optimized $N \times N$ matrix deduplication logic to accelerate batch scanning over massive case datasets.

MiraclePlus (YC China) | Beijing, China

Feb 2023 – Jun 2023

Product Intern

- **Customer Relationship Management (CRM) Data Integration:** Collaborated with engineering teams to design unified matching rules for CRM platforms, deployed fuzzy matching logic for core business fields to guarantee end-to-end data integration accuracy.
- **Data Tracking & Analytics:** Refactored Urchin Tracking Module (UTM) parameter transmission logic to build full-link cross-channel tracking pipelines; wrote Structured Query Language (SQL) scripts on Superset to build conversion funnel visualization dashboards.
- **Data Structuring via AI:** Participated in Prompt optimization strategies, utilizing ChatGPT's generative capabilities to extract and structure non-standardized information from application forms.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, Java, JavaScript
- **Frameworks & Modeling Paradigms:** PyTorch, TensorFlow, NetLogo, SQL